

PHYS 230 - Electricity and Magnetism

Lecture 27

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

6 Direct-Current Circuits

- $R - C$ Circuits (Continued)

7 Magnetic Field and Magnetic Forces

- Magnetic Field
- Magnetic Field Lines and Magnetic Flux

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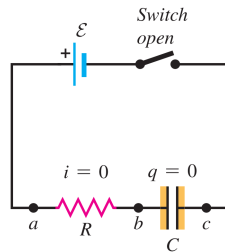
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Exercise I

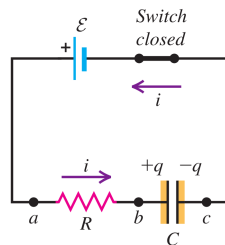
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A $10\text{ M}\Omega$ resistor is connected in series with a $1.0\text{ }\mu\text{F}$ capacitor and a battery with emf 12.0 V . Before the switch is closed at time $t = 0$, the capacitor is uncharged. (a) What is the time constant? (b) What fraction of the final charge Q_f is on the capacitor at $t = 46\text{ s}$? (c) What fraction of the initial current I_0 is still flowing at $t = 46\text{ s}$?

(a) Capacitor initially uncharged



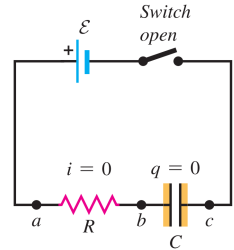
(b) Charging the capacitor



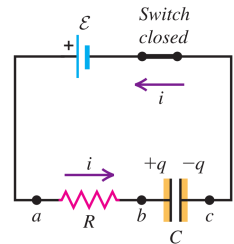
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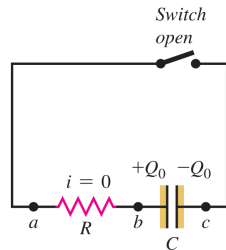


Exercise 2

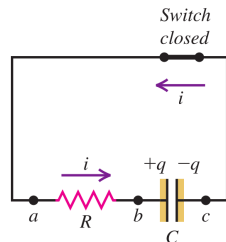
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With the same elements as the previous exercise, in this discharging situation, the capacitor has an initial charge of $5.0 \mu\text{C}$ and is discharged by closing the switch at $t = 0$. (a) At what time will the charge be $0.50 \mu\text{C}$? (b) What is the current at this time?

(a) Capacitor initially charged



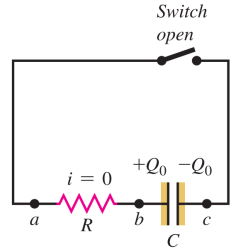
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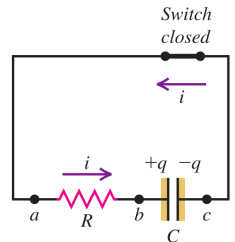
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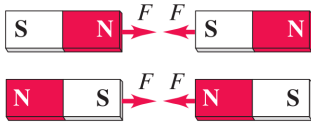
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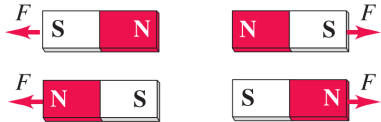
Magnetism

Figure 27.1 (a) Two bar magnets attract when opposite poles (N and S, or S and N) are next to each other. (b) The bar magnets repel when like poles (N and N, or S and S) are next to each other.

(a) Opposite poles attract.



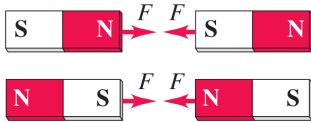
(b) Like poles repel.



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(b) Like poles repel.

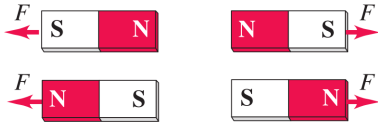
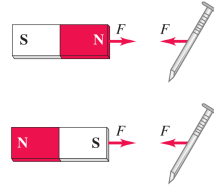
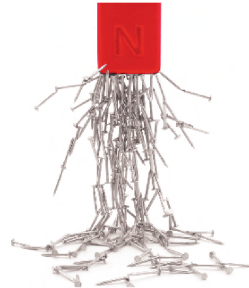


Figure 27.2 (a) Either pole of a bar magnet attracts an unmagnetized object that contains iron, such as a nail. (b) A real-life example of this effect.

(a)

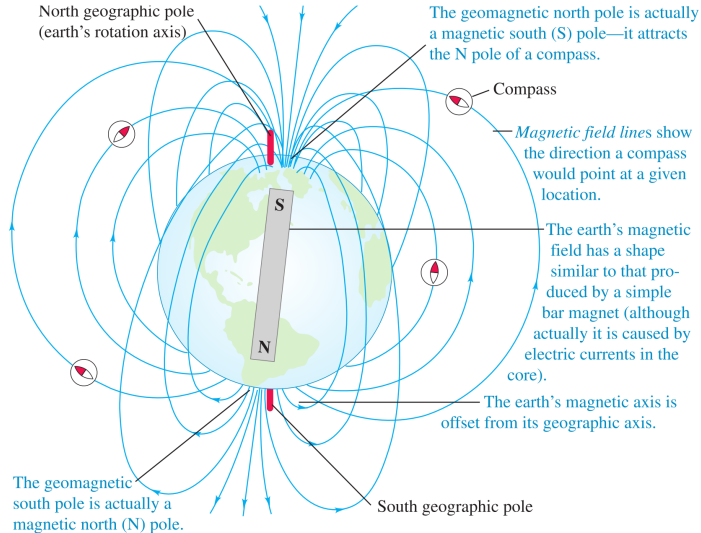


(b)



Magnetism

Figure 27.3 A sketch of the earth's magnetic field. The field, which is caused by currents in the earth's molten core, changes with time; geologic evidence shows that it reverses direction entirely at irregular intervals of 10^4 to 10^6 years.



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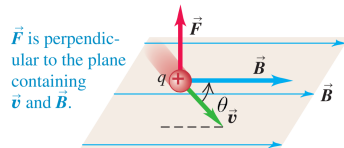
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Force of a Magnetic Field On a Moving Charge

The magnetic force on a charge q moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

A charge moving at an angle θ to a magnetic field experiences a magnetic force with magnitude $F = |q|v_{\perp}B = |q|vB \sin \theta$.



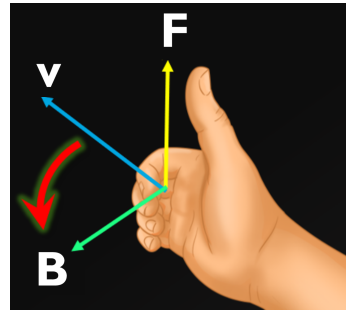
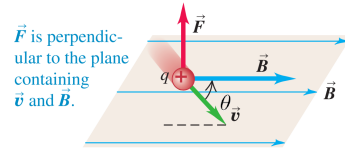
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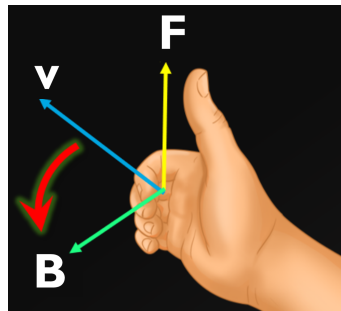
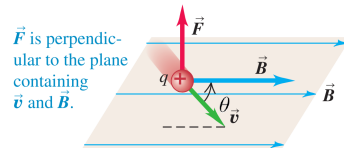
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Direction of $\mathbf{F}$ also depends on the sign of q

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Force of a Magnetic Field On a Moving Charge

Figure 27.7 Finding the direction of the magnetic force on a moving charged particle.

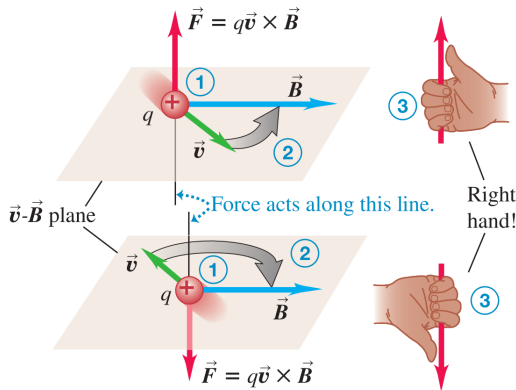
(a)

Right-hand rule for the direction of magnetic force on a **positive** charge moving in a magnetic field:

① Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.

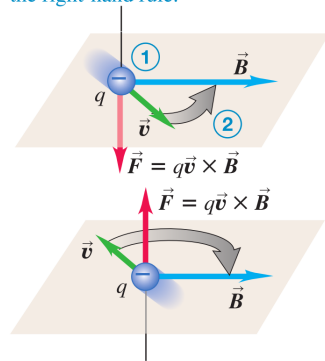
② Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}$ - $\vec{B}$ plane (through the smaller angle).

③ The force acts along a line perpendicular to the $\vec{v}$ - $\vec{B}$ plane. Curl the fingers of your *right hand* around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts.



(b)

If the charge is negative, the direction of the force is **opposite** to that given by the right-hand rule.



Force of a Magnetic Field On a Moving Charge

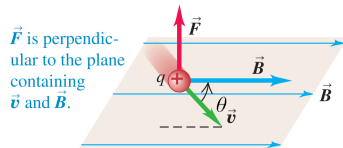
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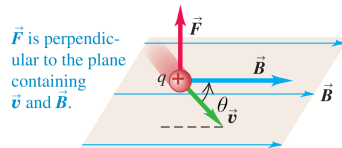
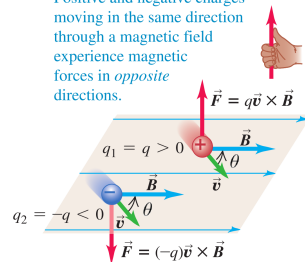


Figure 27.8 Reversing the sign of a moving charge reverses the direction of the magnetic force that acts on it.

Positive and negative charges moving in the same direction through a magnetic field experience magnetic forces in *opposite* directions.



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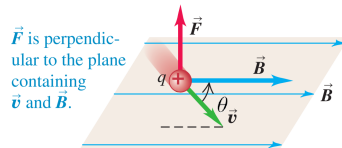
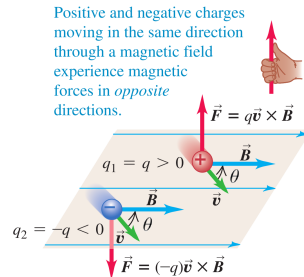


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Units of $\mathbf{B}$ is the **tesla T**,

$$[\mathbf{F}] = [q] [\mathbf{v}] [\mathbf{B}] \Rightarrow [\mathbf{B}] = \frac{[\mathbf{F}]}{[q] [\mathbf{v}]} = \frac{\text{N}}{\text{C} \cdot \text{m/s}} = \frac{\text{N}}{\text{A} \cdot \text{m}} = \text{T}$$

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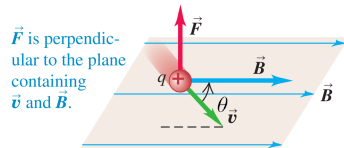
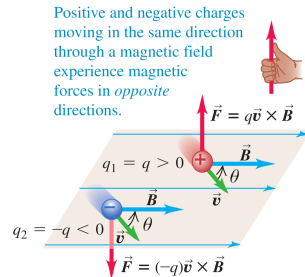


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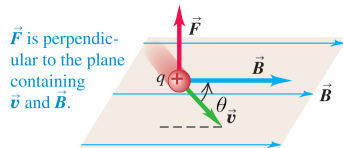
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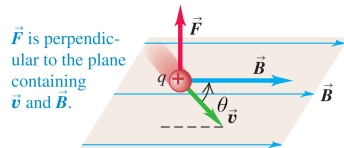
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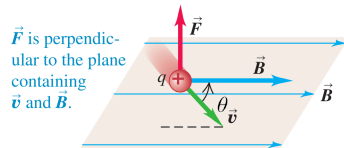
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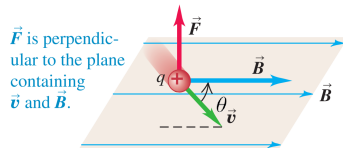
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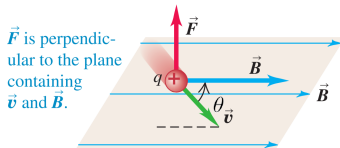
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The total force $\mathbf{F}$ on a charge q moving with velocity $\mathbf{v}$ in an electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ and is

$$\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B} \Rightarrow \boxed{\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})}$$

which is called the **Lorentz Force**

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Exercise 3

<https://epoll.srv.ualberta.ca> – Code: MBD

A beam of protons with $q = e = 1.6 \times 10^{-19} \text{ C}$ moves at $3 \times 10^5 \text{ m/s}$ through a uniform 2.0 T magnetic field directed along the positive z -axis as in figure. The velocity of each proton lies in the xz -plane and is directed at 30° to the $+z$ -axis. Find the force vector on a proton.

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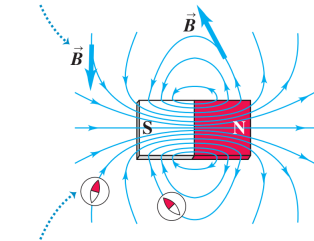
Magnetic Field Lines

Just as in the case of electric field, the magnetic field can also be represented by magnetic field lines

Figure 27.11 Magnetic field lines of a permanent magnet. Note that the field lines pass through the interior of the magnet.

At each point, the field line is tangent to the magnetic-field vector $\vec{B}$.

The more densely the field lines are packed, the stronger the field is at that point.



At each point, the field lines point in the same direction a compass would ...

... therefore, magnetic field lines point away from N poles and toward S poles.

Magnetic Field Lines

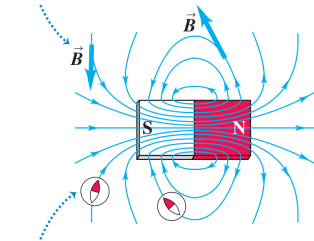
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The direction of $\vec{B}$ at any point is tangent to the field lines, from N to S

Figure 27.11 Magnetic field lines of a permanent magnet. Note that the field lines pass through the interior of the magnet.

At each point, the field line is tangent to the magnetic-field vector $\vec{B}$.

The more densely the field lines are packed, the stronger the field is at that point.



At each point, the field lines point in the same direction a compass would ...

... therefore, magnetic field lines point away from N poles and toward S poles.

Magnetic Field Lines

Just as in the case of electric field, the magnetic field can also be represented by magnetic field lines

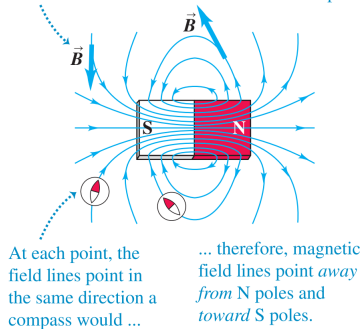
The direction of $\mathbf{B}$ at any point is tangent to the field lines, from N to S

The magnitude $\mathbf{B}$ is larger where lines are closer

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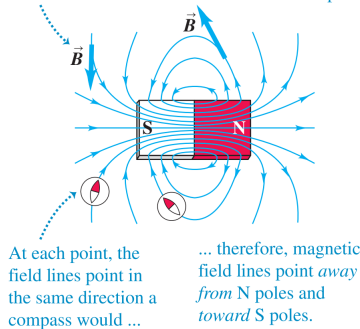
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But unlike electric field lines, magnetic field lines are not lines of force: direction of magnetic force $\mathbf{F}$ is not tangent to the lines of $\mathbf{B}$

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Magnetic Field Lines

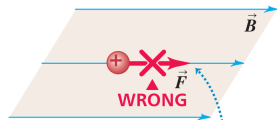
Just as in the case of electric field, the magnetic field can also be represented by magnetic field lines

The direction of $\mathbf{B}$ at any point is tangent to the field lines, from N to S

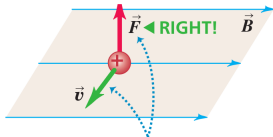
The magnitude $\mathbf{B}$ is larger where lines are closer

But unlike electric field lines, magnetic field lines are not lines of force:
direction of magnetic force $\mathbf{F}$ is not tangent to the lines of $\mathbf{B}$

Figure 27.12 Magnetic field lines are *not* “lines of force.”



Magnetic field lines are *not* “lines of force.”
The force on a charged particle is not along the direction of a field line.

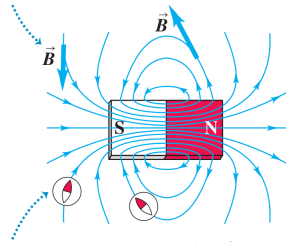


The direction of the magnetic force depends on the velocity $\vec{v}$, as expressed by the magnetic force law $\vec{F} = q\vec{v} \times \vec{B}$.

Figure 27.11 Magnetic field lines of a permanent magnet. Note that the field lines pass through the interior of the magnet.

At each point, the field line is tangent to the magnetic-field vector $\vec{B}$.

The more densely the field lines are packed, the stronger the field is at that point.



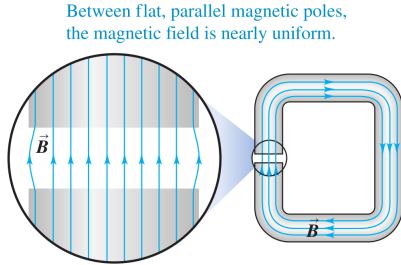
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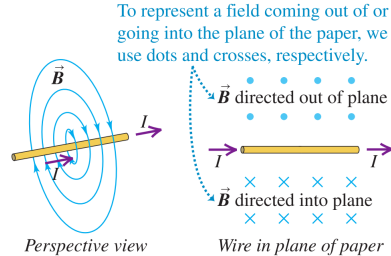
Magnetic Field Lines

Figure 27.13 Magnetic field lines produced by some common sources of magnetic field.

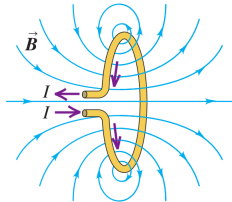
(a) Magnetic field of a C-shaped magnet



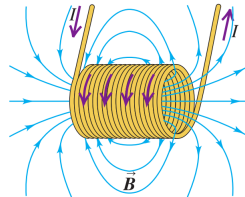
(b) Magnetic field of a straight current-carrying wire



(c) Magnetic fields of a current-carrying loop and a current-carrying coil (solenoid)



Notice that the field of the loop and, especially, that of the coil look like the field of a bar magnet (see Fig. 27.11).

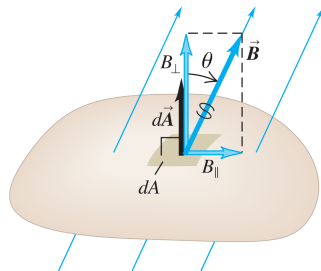


Magnetic Flux

Infinitesimal magnetic flux $d\phi_B$ of a magnetic field $\mathbf{B}$ through an infinitesimal area dA is

$$d\phi_B = \mathbf{B} \cdot d\mathbf{A} = B dA \cos(\theta_{\mathbf{B}, d\mathbf{A}})$$

Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.



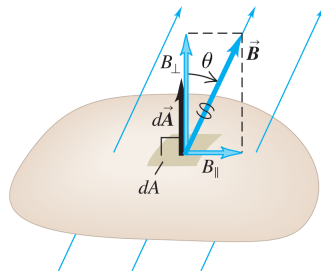
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If the area belongs to a closed surface, then $d\mathbf{A}$ is in the direction of the unit vector $\mathbf{n}$ pointing outwards; for open surfaces you choose the direction

Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.



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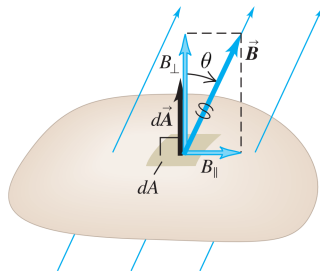
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If the area belongs to a closed surface, then $d\mathbf{A}$ is in the direction of the unit vector $\mathbf{n}$ pointing outwards; for open surfaces you choose the direction

The total magnetic flux is then

$$\phi_B = \int d\phi_B \Rightarrow \phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

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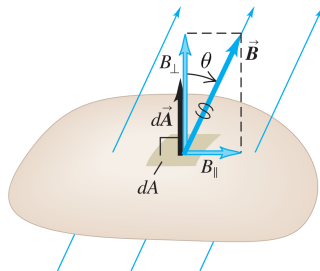
The total magnetic flux is then

$$\phi_B = \int d\phi_B \Rightarrow \phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

SI units of ϕ_B is **weber**, Wb

$$[\phi_B] = [\mathbf{B}] [\mathbf{A}] = \text{T} \cdot \text{m}^2 = \text{Wb}$$

Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.

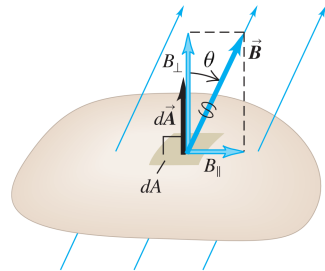


Gauss's Law For Magnetism

Because there is no single magnetic charge (monopole) in nature, Gauss's law for magnetic becomes

$$\oint \mathbf{B} \cdot d\mathbf{A} = 0$$

Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.

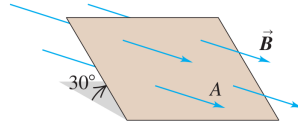


Exercise 1

<https://epoll.srv.ualberta.ca> – Code: MBD

Figure shows a perspective view of a flat surface with area 3.0 cm^2 in a uniform magnetic field $\mathbf{B}$. The magnetic flux through this surface is $+0.90 \text{ mWb}$. Find the magnitude of the magnetic field and the direction of the area vector $\mathbf{A}$.

(a) Perspective view



(b) Our sketch of the problem
(edge-on view)

